

## Ex 05 ARDUINO BASED MANUAL ELECTRONIC COUNTER USING PROTEUS

### Code

```
int x0,x1,x2,x3,x4,x5,x6,x7,x8,x9;
int delay_time=200;
void setup() {
    //configure pin2 as an input and enable the internal pull-up resistor
    pinMode(12, INPUT_PULLUP);
    pinMode(1,OUTPUT); pinMode(2,OUTPUT);
    pinMode(3,OUTPUT);
    pinMode(4,OUTPUT);
    pinMode(5,OUTPUT);
    pinMode(6,OUTPUT);
    pinMode(7,OUTPUT);
}
void loop() {
    //read the pushbutton value into a variable
    int sensorVal = digitalRead(12);
    if (sensorVal == LOW) {
        x0=true;
    }
    while(x0){
        zero();
        sensorVal = digitalRead(12);
        if (sensorVal == LOW) {
            x1=true;
            x0=false;
        }
    }
    while(x1){
        one();
        sensorVal = digitalRead(12);
        if (sensorVal == LOW) {
            x2=true;
            x1=false;
        }
    }
    while(x2){
        two();
        sensorVal = digitalRead(12);
        if (sensorVal == LOW) {
            x3=true;
            x2=false;
        }
    }
```

```
}
while(x3){
three();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x4=true;
x3=false;
}
}
while(x4){
four();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x5=true;
x4=false;
}
}
while(x5){
five();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x6=true;
x5=false;
}
}
while(x6){
six();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x7=true;
x6=false;
}
}
while(x7){
seven();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x8=true;
x7=false;
}
}
while(x8){
eight();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
```

```

x9=true;
x8=false;
}
}
while(x9){
  nine();
  sensorVal = digitalRead(12);
if (sensorVal == LOW) {
  x0=true;
  x9=false;
}
}

}
void zero()
{
  for(int i=1;i<7;i++)
  {
    digitalWrite(i,HIGH);
    digitalWrite(7,LOW);
  }
  delay(delay_time);
}
void one()
{
  digitalWrite(1,LOW);
  digitalWrite(2,HIGH);
  digitalWrite(3,HIGH);
  digitalWrite(4,LOW);
  digitalWrite(5,LOW);
  digitalWrite(6,LOW);
  digitalWrite(7,LOW);
  delay(delay_time);
}
void two()
{
  digitalWrite(1,HIGH);
  digitalWrite(2,HIGH);
  digitalWrite(3,LOW);
  digitalWrite(4,HIGH);
  digitalWrite(5,HIGH);
  digitalWrite(6,LOW);
  digitalWrite(7,HIGH);
  delay(delay_time);
}

```

```
void three()
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,LOW);
digitalWrite(6,LOW);
digitalWrite(7,HIGH);
delay(delay_time);
}

void four()
{
digitalWrite(1,LOW);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,LOW);
digitalWrite(5,LOW);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}
void five()
{
digitalWrite(1,HIGH);
digitalWrite(2,LOW);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,LOW);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}
void six()
{
digitalWrite(1,HIGH);
digitalWrite(2,LOW);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,HIGH);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}void seven()
```

```
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,LOW);
digitalWrite(5,LOW);
digitalWrite(6,LOW);
digitalWrite(7,LOW);
delay(delay_time);
}
void eight()
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,HIGH);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}
void nine()
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,LOW);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}
```

